

3(a)(i)	any four from: (morning / opening / width increasing, context) 1 width increases from 1 to 6.5 mm between 06.00 and 11.00 ; 2 so (more) carbon dioxide can, enter / diffuse in ; 3a high / rising, light intensity → high / more, photosynthesis ; (afternoon / night / closing / width decreasing, context) 3b low / decreasing, light intensity → low / less, photosynthesis ; 4 width decreases from 6.5 to 3 mm between 11.00 and 17.00 ; 5 to, decrease / limit, transpiration / water loss ;	4 IGNORE refs. to K⁺ ions ALLOW mp 3 either in context of 3a or 3b but not twice 1 ALLOW 6.9 for 6.5 mm 3b ALLOW in dark / at night / before 06.00, no photosynthesis 4 ECF from mp 1 if peak time or width wrong 5 CON prevent / stop, for 'decrease'
3(a)(ii)	any three from: 1 K⁺ ions enter (guard) cell ; 2 water enters cell, by osmosis / down water potential gradient → increase in volume / increase in turgor / cell becomes turgid ; 3 action in mp 1 or 2 causally linked to: widens aperture / opens stoma(ta) ;	3 ALLOW ORA throughout 1 CON stoma for 'cell' 2 CON stoma for 'cell' but ECF from mp 1 3 ALLOW opens for 'widens' 3 CON guard cell for 'stoma' 3 IGNORE statement of correlation
3(b)	any three from: <i>ABA concentration is more likely to increase when light intensity is <u>decreasing</u> because:</i> 1 ABA stops H⁺ moving, out of cell / into cell wall ; 2 (ABA) causes Ca²⁺ ions to enter, cytoplasm / cell ; 3 (ABA) causes K⁺ to leave (cytoplasm / cell) ; 4 (ABA) leads to, water leaving (cell) / guard cell becoming flaccid / stomatal closing ;	3 If answer is about when ABA will decrease (not answering question), CON 1 mark then apply ECF after first error If answer states that ABA rises when light intensity is increasing, CON 1 mark then apply ECF after first error 3 IGNORE prevent K⁺ from entering 4 DO NOT ALLOW guard cell closing 4 IGNORE prevent stomata from opening